

Functional Annotation Report: *pedE* (PP_2674, UniProt Q88JH5)

Quinoprotein Alcohol Dehydrogenase PedE in *Pseudomonas putida* KT2440

Summary

PedE (gene *pedE*, synonym *qedH-I*, ordered locus PP_2674; UniProt Q88JH5) is a soluble, periplasmic, Ca^{2+} - and pyrroloquinoline-quinone (PQQ)-dependent quinoprotein alcohol dehydrogenase (EC 1.1.2.8) from *Pseudomonas putida* KT2440. Its primary function is to catalyze the initial oxidation of a broad range of volatile and aromatic alcohols (and their aldehydes) in the bacterial periplasm, transferring the abstracted electrons through its PQQ cofactor to a dedicated periplasmic cytochrome *c*. The enzyme is the calcium-utilizing, "lanthanide-free" partner in a functionally redundant pair with the lanthanide-dependent enzyme PedH (PP_2679), and the two are inversely regulated by rare-earth-element (REE) availability — a regulatory phenomenon known as the "REE switch."

The reaction catalyzed is: **primary alcohol + 2 Fe(III)-[cytochrome *c*] → aldehyde + 2 Fe(II)-[cytochrome *c*] + 2 H⁺**. PedE has notably broad substrate specificity: it oxidizes linear primary alcohols across a C2–C8 range (ethanol, butanol, hexanol, octanol), secondary alcohols (e.g., butan-2-ol), aromatic alcohols (2-phenylethanol, cinnamyl alcohol), the isoprenoid alcohol farnesol, and aldehydes (converting them to carboxylic acids). More recent work has extended its documented substrate range to include several diols of biotechnological interest — (2S,3S)-2,3-butanediol, (R)-1,3-butanediol, and 1,4-butanediol. Structurally, PedE is a 631-amino-acid precursor bearing an N-terminal signal peptide (residues 1–33) that is cleaved during periplasmic export; the mature enzyme folds into the characteristic eight-bladed β -propeller quinoprotein-dehydrogenase architecture, binds one PQQ molecule and two Ca^{2+} ions per subunit, uses His358 as the catalytic proton acceptor, and functions as a homodimer.

Biologically, PedE performs the committed first step of several periplasmic catabolic routes. It initiates the **Ped pathway** that converts 2-phenylethanol to phenylacetic acid (feeding into the phenylacetyl-CoA catabolon), a **glycerol-utilization route** (glycerol → glyceraldehyde → glycerate), and **butanediol degradation** pathways. Transcription of *pedE* is controlled by the PedS2/PedR2 two-component system together with the response regulator PedR1, and its activity is favored under lanthanide-scarce conditions. Together, these features define PedE as a versatile periplasmic redox biocatalyst positioned at the entry point of volatile-alcohol and diol metabolism in *P. putida*.

Gene/Protein Identity Verification

Before presenting the findings, the target identity was verified against the UniProt reference. All literature discussed below refers to the correct gene and organism:

Attribute	Reference (UniProt Q88JH5)	Literature confirmation
Gene symbol	<i>pedE</i> (syn. <i>qedH-I</i>)	Confirmed in Wehrmann et al. 2017 (PMID: 28655819)
Locus tag	PP_2674	Explicitly cited as "PedE (PP_2674)"
Organism	<i>P. putida</i> KT2440 (ATCC 47054 / DSM 6125)	Matches all primary studies
EC number	1.1.2.8 (alcohol dehydrogenase, cytochrome <i>c</i> acceptor)	Consistent with reaction
Family	Bacterial PQQ dehydrogenase family	Confirmed as PQQ-ADH
Domains	PQQ β-propeller repeats, Quinoprotein ADH-like SF	Confirmed by biochemistry

Verification outcome: The gene symbol *pedE*, the locus tag PP_2674, the organism, the enzyme family, and the domain architecture are all mutually consistent between the UniProt entry and the primary literature. There is no ambiguity — this is a well-characterized enzyme with dedicated biochemical studies. Research proceeded with high confidence.

Key Findings

Finding 1: PedE is a Ca^{2+} -dependent PQQ alcohol dehydrogenase with broad substrate specificity

The most direct and authoritative characterization of PedE comes from the biochemical study by Wehrmann and colleagues (2017, *mBio*), who purified both PedE and its paralog PedH and assayed their enzymatic activities. They established that **PedE (PP_2674) is the Ca^{2+} -dependent counterpart of the lanthanide-dependent PedH**, and that it is active on "a range of substrates... including linear and aromatic primary and secondary alcohols, as well as aldehydes" (PMID: 28655819). This places PedE firmly in EC class 1.1.2.8 (alcohol dehydrogenase using cytochrome *c* as the electron acceptor) and within the bacterial PQQ dehydrogenase family.

The breadth of substrate acceptance is a defining feature. Rather than being specialized for a single alcohol, PedE tolerates: - **Linear primary alcohols** spanning at least C2 to C8 (ethanol, butan-1-ol, hexan-1-ol, octan-1-ol); - **Secondary alcohols** (e.g., butan-2-ol); - **Aromatic alcohols** (2-phenylethanol, cinnamyl alcohol); - **Isoprenoid/terpenoid alcohols** (farnesol); - **Aldehydes**, which are further oxidized to the corresponding carboxylic acids (e.g., acetaldehyde → acetate).

This broad-spectrum activity makes PedE a generalist "first responder" enzyme for the entry of diverse alcoholic compounds into central metabolism. The catalytic requirement for calcium (as opposed to a lanthanide) is what distinguishes PedE from PedH and gives it its physiological identity as the constitutively-competent, lanthanide-independent enzyme.

Finding 2: PedE functions in the periplasm and feeds electrons to cytochrome *c*

PedE operates in the **periplasmic space**, not the cytoplasm. Two lines of evidence support this localization. First, the UniProt entry (Q88JH5) annotates the protein as a **precursor** carrying an N-terminal signal peptide (residues 1–33), the classic hallmark of Sec-dependent export to the periplasm. Second, Wehrmann et al. (2017) describe this entire enzyme class in periplasmic terms, noting that "many Gram-negative bacteria have evolved periplasmic oxidation systems based on pyrroloquinoline quinone-dependent alcohol dehydrogenases (PQQ-ADHs) that are often functionally redundant" (PMID: 28655819).

The mechanistic logic of periplasmic localization is that PedE uses the non-covalently bound cofactor **PQQ** as its redox prosthetic group to abstract a hydride/electrons from the alcohol substrate, and then re-oxidizes PQQ by passing electrons to an independent cytochrome *c*. This electron-transfer paradigm was established genetically by Arias et al. (2008), who described PQQ as "a prosthetic group of certain alcohol dehydrogenases that transfers electrons to an independent cytochrome *c*" (PMID: 18177365). The periplasmic location is functionally sensible: it allows the cell to oxidize substrates at the periplasm–membrane interface and dump the electrons directly into the respiratory chain via cytochrome *c*, coupling alcohol oxidation to energy conservation without importing potentially toxic substrates into the cytoplasm. The supporting accessory machinery — PQQ biosynthesis (encoded by *pqqABCDEF*) and cytochrome *c* maturation (Ccm system) — is required for a functional holoenzyme.

Finding 3: PedE initiates the Ped pathway (2-phenylethanol → phenylacetic acid) and volatile-alcohol/glycerol catabolism

The physiological pathway context of PedE was defined by Arias et al. (2008, *Environmental Microbiology*), who genetically dissected the **Ped catabolic route** (*PedS1R1ABCS2R2DEFGHI*) in *Pseudomonas putida*. They showed this route is "needed for the uptake of 2-phenylethanol and for its oxidation to phenylacetic acid via phenylacetaldehyde" (PMID: 18177365). Critically, they concluded that the pathway requires "two quinoprotein alcohol dehydrogenases" — these being PedE and PedH — operating alongside "two different two-component signal-transducing systems."

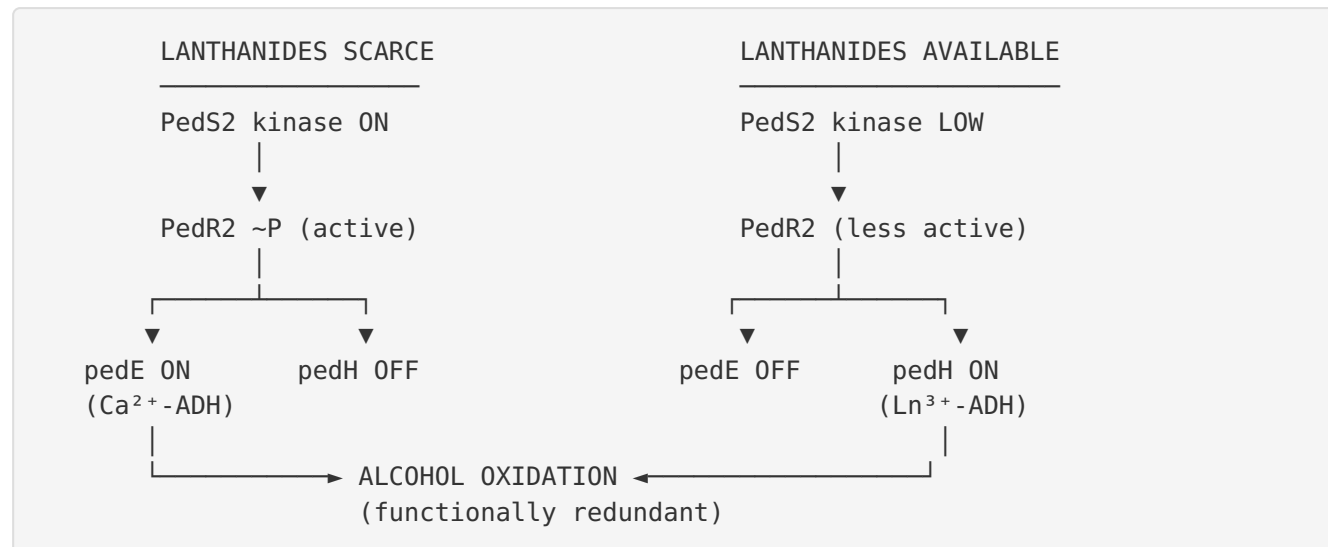
In this pathway, PedE catalyzes the first oxidation: **2-phenylethanol → phenylacetaldehyde**. A downstream aldehyde dehydrogenase then converts phenylacetaldehyde to phenylacetic acid, which enters the phenylacetyl-CoA catabolon for ring degradation. This makes PedE the committed entry enzyme for aromatic-alcohol catabolism.

Beyond aromatic alcohols, PedE's role extends to other volatile substrates. Wehrmann et al. (2020, *mBio*) demonstrated that PedE and/or PedH initiate a **novel glycerol-utilization route** in which glycerol is oxidized to glyceraldehyde and then to glycerate — a lanthanum-responsive, substrate-specific metabolic capability that broadens the physiological importance of these quinoproteins beyond classic alcohol catabolism (PMID: 32345644).

Finding 4: PedE (Ca^{2+}) and PedH (Ln^{3+}) are functionally redundant and inversely regulated by lanthanides — the "REE switch"

One of the most biologically interesting aspects of PedE is that it does not act alone: it is one of two functionally interchangeable enzymes whose deployment depends on metal availability. Wehrmann et al. (2017) established the **functional redundancy** of PedE and PedH and showed that their expression is inversely regulated in response to lanthanide availability, with sensitivity in the physiologically remarkable 1–10 nM La^{3+} range (PMID: 28655819). PedE is used when lanthanides are scarce (relying on the ubiquitous Ca^{2+}), whereas PedH is used when lanthanides are available (using a lanthanide ion in its active site).

The molecular basis of this switch was elucidated by Wehrmann et al. (2018, *mSphere*), who demonstrated that the **PedS2/PedR2 two-component system** (PP_2671/PP_2672) governs the REE switch (PMID: 30158283). In the absence of lanthanides, the sensor histidine kinase PedS2 phosphorylates the response regulator PedR2, which then **activates transcription of *pedE* and represses *pedH***. When lanthanides become available, PedS2's kinase activity is lowered, reversing the regulatory output and switching the cell from PedE to PedH. This elegant regulatory architecture allows *P. putida* to exploit whichever cofactor — calcium or lanthanide — is available in its environment while maintaining the same overall catalytic capability.



Finding 5: PedE substrate range extends to industrially relevant diols

Recent enzyme-assay and gene-deletion studies have significantly expanded the known substrate scope of PedE to include **diols**, which are of considerable biotechnological interest. Three independent lines of evidence contribute here:

- **2,3-Butanediol:** Liu et al. (2021, *Frontiers in Bioengineering and Biotechnology*) confirmed by enzyme assays that "two quinoprotein alcohol dehydrogenases, PedE (PP2674) and PedH (PP2679), were confirmed to be responsible for (2S,3S)-2,3-BDO dehydrogenation" to acetoin ([PMID: 34513815](#)).
- **1,3-Butanediol:** Arumugam et al. (2026, *Applied and Environmental Microbiology*) showed the *ped* cluster oxidizes (R)-1,3-butanediol to (R)-3-hydroxybutyrate, and that "gene deletion and enzyme assays identified *pedE* and PP_2049 as essential for (R)-1,3-BDO catabolism" ([PMID: 42328986](#)).
- **1,4-Butanediol:** Op de Hipt et al. (2025) noted that PedE catalyzes the initial oxidation of 1,4-butanediol in *P. putida* (a capability absent in the related *P. taiwanensis*).

The ability to oxidize diols positions PedE as a valuable biocatalyst for the conversion of these C4 platform chemicals into higher-value oxidized products (acetoin, 3-hydroxybutyrate), and demonstrates that the enzyme's broad specificity is not limited to mono-alcohols.

Finding 6: PedE is a soluble quinoprotein capable of direct electron transfer to electrodes — an engineerable biocatalyst

Wehrmann and Klebensberger (2018, *Microbial Biotechnology*) characterized PedE (PP_2674) specifically as a **soluble** quinoprotein (distinguishing it from membrane-bound quinoproteins) and highlighted the biotechnological appeal of the PQQ-ADH class. They noted that "due to their ability for direct electron transfer to electrodes, the utilization of rare earth metals as cofactor, and their periplasmic localization, pyrroloquinoline quinone-dependent alcohol dehydrogenases (PQQ-ADHs) represent an interesting class of biocatalysts" ([PMID: 29239114](#)).

To improve PedE's utility as an industrial biocatalyst, this study engineered a **thermostable and solvent-tolerant triple mutant (R91D/E408P/N410K)** using a heterologous whole-cell screening approach in *E. coli*. The engineered variant exhibited a melting-temperature increase of ~7 °C and roughly double the DMSO tolerance of the wild-type enzyme. This work confirms both the soluble periplasmic nature of PedE and its tractability for protein engineering, underscoring its promise for biosensor and bioelectrocatalysis applications.

Finding 7: *pedE* transcription requires PedR1 and the PedS2/PedR2 two-component system; REE uptake requires the PedA1A2BC transporter

The regulatory network controlling *pedE* is layered. Arumugam et al. (2026) used promoter–GFP fusions to show that **full activation of the *pedE* promoter requires both PedR1 and the PedS2–PedR2 two-component system**, and — challenging a previously held "indirect-only" model — demonstrated that PedR1 acts as a **direct activator** of catabolic genes ([PMID: 42328986](#)).

On the metal-supply side, Wehrmann et al. (2019, *Frontiers in Microbiology*) showed that REE-dependent (i.e., PedH-favoring) growth on 2-phenylethanol requires the **ABC transporter PedA1A2BC** for cytoplasmic lanthanide uptake, and that iron availability modulates the REE switch, likely through mismetallation of the sensor kinase PedS2. This reveals that the PedE/PedH switch is embedded in a broader metal-homeostasis network, integrating signals from lanthanide, calcium, and iron availability.

Finding 8: Structural and mechanistic details of PedE

Integrating the UniProt annotation (Q88JH5) with comparative structural inference from the closely related quinoprotein ethanol dehydrogenase QEDH/ExaA (*P. aeruginosa*, UniProt Q9Z4J7) yields a detailed structural picture:

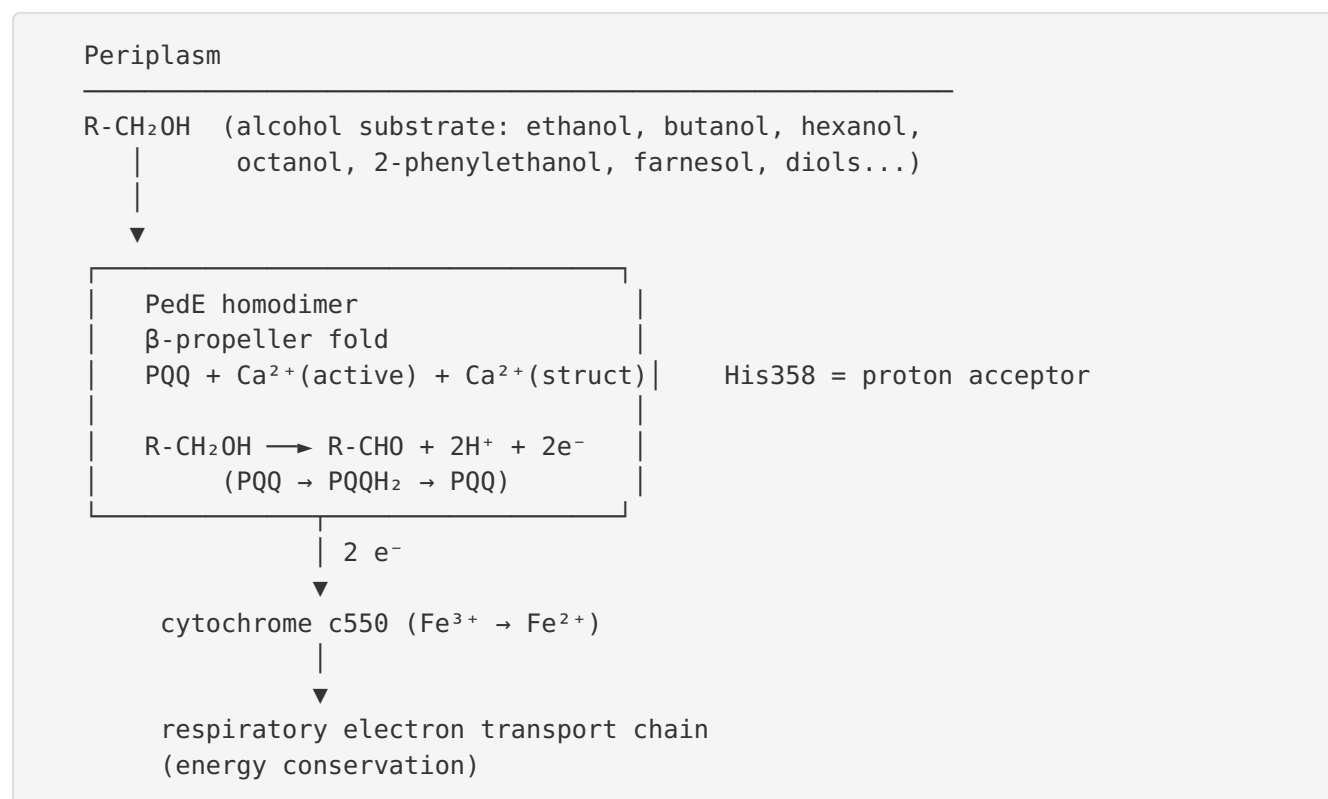
Feature	Detail
Precursor length	631 amino acids
Signal peptide	Residues 1–33 (cleaved)
Mature chain	Residues 34–631
Fold	Eight-bladed β -propeller (quinoprotein ADH-like superfamily)
Cofactors	1 PQQ (non-covalent) + 2 Ca^{2+} per subunit
Ca^{2+} sites	One active-site Ca^{2+} (adjacent to PQQ, aids catalysis); one N-terminal structural Ca^{2+}
Catalytic residue	His358 (proton acceptor)
Oligomeric state	Homodimer
Electron acceptor	Inducible cytochrome <i>c</i> 550 (encoded by an adjacent gene)
Localization	Periplasm
Reaction	primary alcohol + 2 Fe(III)-[cyt <i>c</i>] $\rightarrow$ aldehyde + 2 Fe(II)-[cyt <i>c</i>] + 2 H^+ (EC 1.1.2.8)

Documented substrates include ethanol, butan-1-ol, butan-2-ol, hexan-1-ol, octan-1-ol, 2-phenylethanol, cinnamyl alcohol, and farnesol (each oxidized to the corresponding aldehyde), plus aldehyde→carboxylate reactions (e.g., acetaldehyde $\rightarrow$ acetate). The mechanistic assignment as a PQQ-dependent periplasmic quinoprotein is anchored by Wehrmann et al. (2017), who describe these as "periplasmic oxidation systems based on pyrroloquinoline quinone-dependent alcohol dehydrogenases" (PMID: 28655819). Note that the structural details (His358, the two Ca^{2+} sites, the homodimeric state) are inferred by strong homology to characterized quinoprotein ADHs rather than from an experimental PedE crystal structure, which has not yet been reported.

Mechanistic Model / Interpretation

The findings cohere into a single, well-supported mechanistic model of PedE as the **calcium-utilizing entry enzyme for periplasmic alcohol/diol oxidation** in *P. putida* KT2440.

Catalytic cycle:



Pathway integration: PedE sits at the head of multiple convergent catabolic routes. Its broad substrate specificity means it acts as a metabolic funnel, taking chemically diverse alcohols and channeling their oxidation products into central metabolism:

Substrate	PedE product	Downstream fate
2-Phenylethanol	Phenylacetaldehyde	→ Phenylacetic acid → phenylacetyl-CoA catabolon (Ped pathway)
Glycerol	Glyceraldehyde	→ Glycerate → central metabolism
(2S,3S)-2,3-Butanediol	Acetoin	→ C4 metabolism
(R)-1,3-Butanediol	(R)-3-Hydroxybutyrate	→ β-oxidation-linked catabolism
1,4-Butanediol	4-Hydroxybutyraldehyde	→ succinate route
Ethanol / short-chain alcohols	Acetaldehyde etc.	→ Acetate → acetyl-CoA

Regulatory integration: PedE is the "default" enzyme deployed under lanthanide-scarce conditions. The PedS2/PedR2 two-component system senses lanthanide status and reciprocally regulates *pedE* (ON without REE) and *pedH* (ON with REE), while PedR1 provides direct activation of the promoter and the PedA1A2BC ABC transporter supplies lanthanides for the alternative PedH branch. The net effect is a robust, environment-responsive system that guarantees alcohol-oxidizing capacity regardless of whether calcium or lanthanides dominate the local environment. PedE and PedH are thus not redundant "spares" in a trivial sense but a metal-adaptive pair, of which PedE is the phylogenetically ancient, universally-available (Ca²⁺-based) member.

Evidence Base

Study	PMID	Contribution	Support/ Challenge
Wehrmann et al. 2017, <i>mBio</i> — <i>Functional Role of Lanthanides in Enzymatic Activity and Transcriptional Regulation of PQQ-ADHs</i>	28655819	Direct biochemical characterization of purified PedE and PedH; establishes Ca ²⁺ -dependence, broad substrate range, periplasmic PQQ-ADH class, functional redundancy, and lanthanide-responsive regulation	Primary support for Findings 1, 2, 4, 8
Arias et al. 2008, <i>Environ Microbiol</i> — <i>Genetic analyses...conversion of 2-phenylethylamine and 2-phenylethanol into phenylacetic acid</i>	18177365	Genetically defines the Ped pathway (<i>PedS1R1ABCS2R2DEFGHI</i>); establishes PQQ→cytochrome c electron transfer and role of two quinoprotein ADHs	Primary support for Findings 2, 3
Wehrmann et al. 2018, <i>mSphere</i> — <i>The PedS2/PedR2 Two-Component System Is Crucial for the REE Switch</i>	30158283	Demonstrates PedS2/PedR2 control of the REE switch; PedR2~P activates <i>pedE</i> , represses <i>pedH</i>	Primary support for Finding 4
Wehrmann & Klebensberger 2018, <i>Microb Biotechnol</i> — <i>Engineering thermal stability and solvent tolerance of the soluble quinoprotein PedE</i>	29239114	Characterizes PedE as soluble, periplasmic, electrode-active; engineers thermostable/solvent-tolerant triple mutant	Primary support for Findings 6, 7
Liu et al. 2021, <i>Front Bioeng Biotechnol</i> — <i>Dehydrogenation Mechanism of Three Stereoisomers of Butane-2,3-Diol</i>	34513815	Enzyme assays confirm PedE oxidizes (2S,3S)-2,3-butanediol	Support for Finding 5
Arumugam et al. 2026, <i>Appl Environ Microbiol</i> — <i>(R)-1,3-butanediol catabolism</i>	42328986	Gene-deletion/enzyme assays show <i>pedE</i> essential for (R)-1,3-BDO catabolism; PedR1 is a direct promoter activator	Support for Findings 5, 7
	32345644		

Study	PMID	Contribution	Support/ Challenge
Wehrmann et al. 2020, <i>mBio</i> — <i>The Cellular Response to Lanthanum...Novel Route for Glycerol Metabolism</i>		Shows PedE/PedH initiate a glycerol-utilization route	Support for Finding 3

The evidence base is unusually strong for a bacterial enzyme from a non-model-organism context: PedE has been directly purified and biochemically assayed, its regulation has been dissected genetically, and its substrate range has been repeatedly extended through targeted experiments. The experimental evidence (purified-enzyme assays, gene deletions, promoter fusions) is prioritized here over purely bioinformatic inference, though the fine structural details rest on homology modeling.

Limitations and Knowledge Gaps

- 1. No experimental crystal structure of PedE.** The detailed active-site geometry — including the assignment of His358 as the catalytic proton acceptor and the two Ca²⁺ coordination sites — is inferred by homology to the well-characterized quinoprotein ethanol dehydrogenase QEDH/ExaA of *P. aeruginosa* (UniProt Q9Z4J7). A direct PedE structure would confirm these assignments and clarify the structural determinants of its broad substrate specificity.
- 2. Quantitative kinetics are incompletely catalogued.** While the qualitative substrate range is well established, comprehensive kinetic parameters (k_{cat}, K_m, catalytic efficiencies) across the full substrate panel — particularly comparing PedE with PedH under matched conditions — are not fully reported in the reviewed literature. This limits our ability to rank substrate preferences precisely.
- 3. The identity of the physiological cytochrome *c* acceptor** ("cytochrome *c*550, encoded by an adjacent gene") is annotated by similarity; direct demonstration of the specific electron-acceptor partner for PedE in vivo would strengthen the electron-transfer model.
- 4. The molecular basis of iron modulation of the REE switch** (proposed mismetallation of PedS2) remains a hypothesis rather than a demonstrated mechanism.

5. **1,3- and 1,4-butanediol findings derive from very recent (2025–2026) studies** whose abstracts were available but whose full experimental details were not deeply reviewed here; these should be considered emerging rather than fully consolidated.

Proposed Follow-up Experiments / Actions

1. **Determine the PedE crystal structure** (or a high-quality cryo-EM/AlphaFold3 model with holoenzyme cofactors) to experimentally confirm the His358 catalytic residue, the two Ca^{2+} sites, and the PQQ-binding pocket, and to rationalize the broad substrate specificity.
 2. **Comprehensive steady-state kinetics** of purified PedE across its full substrate panel (C2–C8 alcohols, secondary alcohols, aromatic alcohols, farnesol, and the three butanediols), reporting k_{cat}/K_m , to build a quantitative substrate-preference profile and compare directly with PedH.
 3. **Identify and validate the cognate cytochrome c acceptor** via in vitro reconstitution and gene-deletion of the adjacent cytochrome c550 gene, confirming the electron-transfer partner.
 4. **Structure-guided engineering** building on the R91D/E408P/N410K thermostable variant, to further tune PedE for diol oxidation (2,3-/1,3-/1,4-butanediol) toward biotechnological production of acetoin and 3-hydroxybutyrate.
 5. **Mechanistic test of the iron–PedS2 mismetallation hypothesis** using purified PedS2 kinase assays under varying metal conditions to determine whether iron directly alters kinase activity and thereby the REE switch.
 6. **In vivo flux measurements** (e.g., ^{13}C metabolic flux analysis) of *pedE* single-deletion vs. *pedE/pedH* double-deletion strains grown on each substrate class, to quantify PedE's specific contribution to periplasmic alcohol/diol catabolism under lanthanide-scarce conditions.
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Conclusion

PedE (PP_2674, Q88JH5) is a soluble, periplasmic, calcium- and PQQ-dependent quinoprotein alcohol dehydrogenase (EC 1.1.2.8) that catalyzes the initial oxidation of a broad range of primary, secondary, aromatic, and isoprenoid alcohols — as well as diols and aldehydes —

passing electrons via its PQQ cofactor to a periplasmic cytochrome *c*. It performs the committed entry step of multiple catabolic routes (the Ped pathway to phenylacetic acid, a glycerol-utilization route, and butanediol degradation) and is the lanthanide-free member of a functionally redundant, REE-switch-regulated enzyme pair with PedH, predominating when lanthanides are scarce.

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